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A MODEL FOR A MULTI-ITEM, MULTI-ECHELON, MULTI-INDENTURE INVENTORY SYSTEM*†

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The purpose of this paper is to describe a mathematical model, called MOD-METRIC, for the control of a multi-item, multi-echelon, multi-indenture inventory system for recoverable items, that is, items subject to repair when they fail. Discussion is limited to two-echelon multi-item systems in which an item may be demanded at any one of several locations called bases; in turn, these bases receive inventory from a central location called a depot. The objectives of the model are to describe the logistics relationship between an assembly and its subassemblies, and to compute spare stock levels for both echelons for the assembly and subassemblies with explicit consideration of this logistics relationship. In particular, the model is used to determine the base and depot spare stock levels which minimize total expected base backorders for the assembly subject to a system investment constraint. An example is given showing how the model can be used to calculate spare engine and engine module stock levels. MOD-METRIC has been implemented by the Air Force as the method for computing recoverable spare stock levels for the F-15 weapon system.

1. Introduction

During the past two decades a substantial number of mathematical models have been developed relating to various aspects of managing inventory systems. One of these models, called METRIC (Multi-Echelon-Technique-for-Recoverable-Item-Control) [7], was specifically designed for the Air Force at The RAND Corporation as a method for determining stock levels in a two-echelon inventory system—base and depot—for a class of items called recoverable items, that is, items subject to repair when they fail. These recoverable items are typically expensive and their individual demand rates are usually relatively low; however, their proper management is imperative since approximately 65 percent of the Air Force's investment in spares is concentrated in these items.

The purpose of this paper is to describe a mathematical model for the control of a multi-item, multi-echelon, multi-indenture inventory system for recoverable items which the Air Force has recently implemented. This model, called MOD-METRIC, an extension of METRIC, permits the explicit consideration of a hierarchical parts structure. For example, a major assembly may consist of a casing and several components. The components are subordinate to the assembly in the parts hierarchy. The objectives of the model are to describe the logistics relationship between the components and the final assembly, and to compute base and depot spare stock levels for all items with explicit consideration of this logistics relationship.

The next section contains an example of the structure of the Air Force inventory and maintenance systems as it pertains to engines. §3 contains a brief review of RAND's METRIC model. In the fourth section, the MOD-METRIC model is presented followed by a description of an algorithm for determining stock levels in the fifth section. A summary and concluding remarks are given in the final section.

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2. System Structure

In this paper discussion is limited to two-echelon, multi-item systems in which a particular item may be demanded at any one of several locations called bases; in turn, these bases receive their inventory from a central location called a depot. To obtain an understanding of the system's structure and the corresponding inventory stockage problem, let us consider a specific example.

To improve maintainability, aircraft engines for the latest weapon systems are more modularly designed than previous engines. The engines consist of several modules and perhaps a casing and some external parts; however, the vast majority of recoverable parts are located in the modules. The normal maintenance concept is to remove an engine which has failed, replace that engine with a serviceable one, determine the module (or modules) containing the defective part (parts), remove that module from the engine and replace it with a like module from serviceable module stock. Depending on the type of failure, the module will be repaired either at base or depot level. In a small number of cases, the entire engine will be returned to the depot for repair. This generally will occur when more than one module is damaged, for example, if there is extensive foreign object damage; however, the concept is to perform almost all maintenance on modules.

Clearly the objective of this concept is to shorten the repair time for whole engines. The anticipated time in maintenance for the engine will be confined to fault isolation, removing and replacing the defective module, and testing. If the required maintenance takes place at base level, the maintenance flow time on the engine is reduced approximately by the time it takes to repair the damaged module. On the other hand, if the failed module requires depot maintenance, the whole engine is not shipped to the depot, but is returned to a serviceable condition through this modular remove and replace repair concept. This procedure substantially increases the number of days per year that a spare engine will be in a serviceable condition. In addition, if a module or whole engine is to be sent to the depot for repair, a demand for a similar asset is placed on the depot by that base. The resupply time for an item, that is, the time it takes to replace an asset which was demanded from base supply with a serviceable asset, depends on the location at which the necessary maintenance will take place as well as the type of repair that is required. A diagram of the repair and inventory systems for a modularized engine is given in Figure 1.

3. Metric: A Review

The METRIC model is described in detail in reference [7]. The purpose of this section is to provide a brief review of this model. METRIC is a model for determining both requirements and distribution of recoverable items in a two-echelon inventory system. The objective of this model is to determine the base and depot stock levels[1] which minimize total expected base level backorders for a specific set of items and bases subject to an investment constraint. By definition, a backorder exists at a point in time at which there is an unsatisfied demand at base level. If backorder days are accumulated over an extended length of time, the daily expected number of backorder days can be found by dividing the accumulated backorder days by the number of days in the data period and calculating the mathematical expectation of this quantity. No penalty is directly assessed for depot backorders. Depot backorders are considered only insofar as they influence base backorders.

[1] Stock is defined to be the total stock on hand plus on order plus in repair minus backorders.

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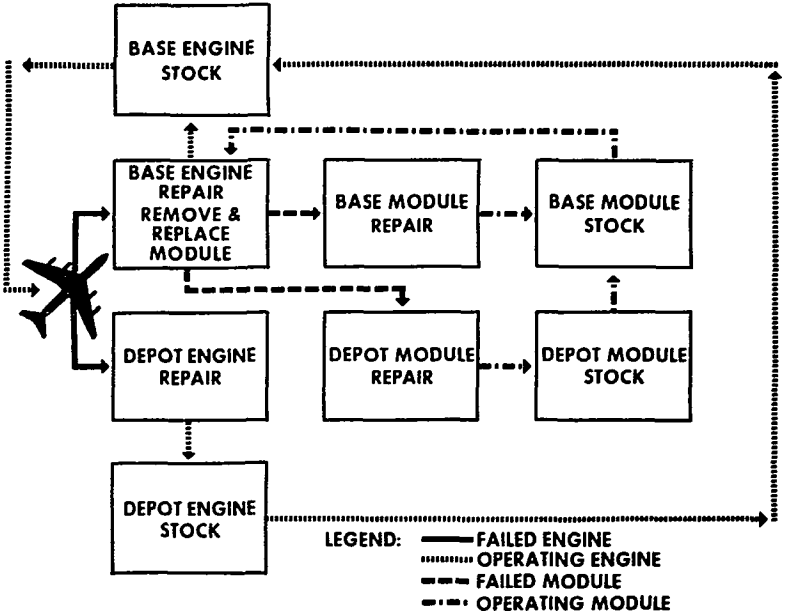

FIGURE 1. Module repair concept.

Various data are required as input parameters to the METRIC model. These are the average base and depot repair times for each item, unit costs, certain probability distribution parameters, Not-Reparable-This-Station (NRTS) rates, and average order and ship times. Minimum and maximum stock levels can be specified. In addition, the following assumptions are made:

1. A stationary compound Poisson probability distribution describes the demand process for each item; predictable nonstationarity of the process can easily be included. 2. There is no lateral resupply between bases. 3. There are no condemnations (all failed parts are repaired). 4. A failure of one type of item is statistically independent of those that occur for any other type of item. 5. Repair times are statistically independent. 6. There is no waiting or batching of items before repair is started on an item (infinite channel queuing assumption). 7. The level at which repair is performed depends only on the complexity of the repair. 8. Items and bases may have different essentialities; however, items are normally considered to be equally essential.

These assumptions are, for the most part, valid for use in a planning model. There is one exception, however. Assumption 8 is inappropriate for modularly designed items. It is in this area where MOD-METRIC differs from METRIC.

4. MOD-METRIC: A Multi-Item, Multi-Echelon, Multi-Indenture Inventory Model

In this section, a model is presented which extends METRIC to include a hierarchical or indentured parts structure. The model permits two levels of parts to be considered, an assembly and its components. It is possible to extend the analysis to additional levels of indenture or to more echelons of supply using a straightforward extension of the methods described below.

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The descriptions of the inventory and maintenance systems given in the previous section are also applicable to MOD-METRIC. In addition, assumptions 2 through 7 stated in that section are appropriate for MOD-METRIC as well. In the remainder of this paper it is assumed for ease of exposition that the demand process is a simple Poisson process. It should be noted that in the model which has been implemented that the demand process is assumed to be described by a Poisson process whose mean is a random variable which is distributed according to a gamma distribution. In addition, a Bayesian procedure is being implemented to continuously update the parameters of the gamma prior distribution. For complete details see [5].

Recall that in METRIC, the objective is to minimize expected base backorders for all items subject to an investment constraint; the MOD-METRIC objective is to minimize the expected base backorders for the end item subject to an investment constraint on the total dollars allocated to the end item and its components. In order to make the discussion more explicit, let us return our attention to the engine-module mix problem. In METRIC a backorder on a module and a backorder for an engine are assumed to be equally undesirable; however, these backorders affect the system in different ways. The maintenance concept depicted in Figure 1 illustrates that engines and modules have different functions in the repair process. When an engine fails on an aircraft it must be replaced with another engine. Consequently an engine backorder indicates that an aircraft is missing an engine and is unavailable to perform its flying mission. Modules, on the other hand, are used only to repair engines. A backorder for a module only delays the repair of an engine. The impact of module backorders and engine backorders is clearly not the same. To compute the effectiveness of the supply system to meet demands for engines, it is necessary to explicitly describe the system's relationship between engines and modules. This relationship is expressed in the equation which represents the average engine resupply time.

The value of T_i, the average engine resupply time at base i, depends on several factors. Recall that the resupply time of an item depends on whether the item is to be repaired at a base or the depot. If it is repaired at base level, the resupply time is the time it takes to flow the unit through the base maintenance system; if the item is repaired at the depot, the resupply time consists of the time to place the depot order for a serviceable part and to receive the part from the depot, assuming a serviceable asset is on hand at the depot. However, when there are no serviceable assets on hand at the depot, an additional delay is included in the resupply time. The expected delay clearly depends on the depot engine stock level. Let us now describe this process in more detail.

The average resupply time, given that the engine is to be repaired at depot level, is the sum of the average order and ship time plus the average delay at the depot due to lack of an available serviceable replacement. Let A_i designate the average order and ship time for an engine at base i. The average depot delay can be developed in the following manner.

Let D represent the average depot repair time (in days). If depot stock is theoretically infinite, the item will experience no delay; however, if depot stock equals zero, the item will, on the average, be delayed D days. For a finite positive depot stock level, the average delay is between 0 and D. To compute the average delay, it is necessary to determine the expected number of depot backorders.

Let s_0 be the depot engine stock, r_i the probability an engine will be repaired at base i, λ_i the daily engine removal rate at base i, and $p(x|\alpha)$ the probability of x units in resupply when α is the customer arrival rate. Define $\lambda = \sum_i (1 - r_i)\lambda_i$.

Thus, the probability distribution of the number of daily depot demands is Poisson

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distributed with mean λ. Then the probability distribution of the number of units in resupply is Poisson with mean λT, where T is the average resupply time [2]. Therefore the expected number of units backordered at the depot is

$$B(s_0|\lambda D) = \sum x > s_0 \, (x - s_0) p(x|\lambda D). \tag{1}$$

This expression can also be interpreted as the expected number of units on which delay is being incurred at a random point in time. Consequently equation (1) has the dimension of delay days per day. Dividing this expression by the expected number of demands per day yields a statistic which has the dimension of delay days per demand. Thus

$$\begin{aligned} \delta(s_0)D &= B(s_0|\lambda D)/\lambda \\ &= \text{Expected Backorders/Expected Daily Demand} \end{aligned}$$

represents the expected depot delay per demand.

Let the average resupply time, given an engine is repaired at base level, be denoted by B_i. Then B_i is equal to the average remove and replace time, given the necessary module is available, plus the expected delay due to the unavailability of the module which is required to repair the engine. Then $B_i = R_i + \Delta_i$, where

R_i = the average repair time at base i if modules are available, and

Δ_i = the average delay in base engine repair due to the unavailability of a needed module.

For convenience, let us assume that if an engine requires repair and that repair is made at base level, the probability that more than one module requires maintenance is zero. In practice, the effect of this assumption is minimal since the probability of joint failures requiring base level repair is quite small. In fact, this assumption produces conservative results, that is, expected backorders are somewhat overstated, although only very slightly. In addition, for simplicity assume that if an engine is repaired at base level, then the failure is located in one of the modules.

Let the expected delay in engine base repair time due to a backorder on module j at base i be represented by Δ_{ij}. Then

$$\Delta_{ij} = \frac{\sum_{x_{ij} > s_{ij}} (x_{ij} - s_{ij}) \; p(x_{ij} \mid \lambda_{ij} T_{ij})}{\lambda_{ij}},$$

where

λ_{ij} = average number of daily removals of module j at base i,

s_{ij} = stock level of module j at base i, and

T_{ij} = average resupply time for module j at base i.

Then $T_{ij} = r_{ij}B_{ij} + (1 - r_{ij})(A_{ij} + \delta_j D_j)$, where

r_{ij} = the probability that a failure isolated to module j will be repaired at base level,

B_{ij} = average base repair time for module j at base i,

A_{ij} = average order and ship time for module j at base i,

D_j = average depot repair time for module j,

$$\delta_j = \frac{\sum_{x > s_{0j}} (x - s_{0j}) p(x \mid \theta_j D_j)}{\theta_j D_j},$$

s_{0j} = the stock level of module j at the depot, and $\theta_j = \sum_{i=1}^{M} \lambda_{ij}(1 - r_{ij})$, where M = the number of bases. The expected delay in engine repair at base i due to modules is

$$\Delta_i = \frac{1}{r_i \lambda_i} \sum_{j=1}^{N} \lambda_{ij} \Delta_{ij},$$

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where

N = number of modules and

λ_i = the removal rate for engines at base i.

Thus we have shown the average resupply time for an engine, T_i, can be represented as $T_i = r_iB_i + (1 - r_i)(A_i + \delta(s_0)D) = r_i(R_i + \Delta_i) + (1 - r_i)(A_i + \delta(S_0)D)$. The objective of MOD-METRIC is to employ this relationship to determine engine and module stock levels. The difference between the METRIC and the MOD-METRIC models is the manner in which this average resupply time is expressed.

5. An Algorithm for Determining Stock Levels

The mathematical statement of the problem to be solved is

$$\min \sum_{i=1}^{M} \sum_{x_i=s_i+1}^{\infty} (x_i - s_i)p(x_i|\lambda_i T_i)$$

subject to

$$\sum_{i=1}^{M} [c_E s_i + \sum_{j=1}^{N} c_j s_{ij}] + \sum_{j=1}^{N} c_j s_{0j} + c_E s_0 \leqq C \tag{2}$$

where

s_i = stock level of spare engines at base i,

c_E = unit cost of an engine (end item),

c_j = unit cost of module j, and

C = dollar budget limit.

Unfortunately this problem is not separable. One way to circumvent this situation is to employ a partitioning procedure in which Problem (2) is broken into two parts; one part corresponding to an optimization problem for engines, and the other pertaining to the modules. The algorithm, in its simplest form, requires the solution of many subproblems each of which corresponds to a particular division of the available investment budget between engines and modules.

A Rudimentary Algorithm for Solving Problem (2)

Step 0: Set a lower bound constraint on investment in both engines and modules. Let f and g represent the minimum engine and module investment levels, respectively. Let $g' = g$ and $z' = \infty$. Define b to be the budget increment.

Step 1: Solve $\min \sum_i r_i\lambda_i\Delta_i$

subject to

$$\sum_{j=1}^{N} (c_j s_{0j} + \sum_{i=1}^{M} c_j s_{ij}) \leqq g' \tag{3}$$

where $s_{ij} = 0, 1, \cdots; i = 0, 1, \cdots, M$; and $j = 1, \cdots N$.

Step 2: Solve $\min z = \sum_{i=1}^{M} \sum_{x_i>s_i} (x_i - s_i)p(x_i|\lambda_i T_i)$

subject to

$$\sum_{i=1}^{M} c_E s_i + c_E s_0 \leqq C - g' \tag{4}$$

where T_i is calculated using the module stock levels determined in Step 1 and $s_i = 0, 1, \cdots; i = 0, \cdots, M$.

Step 3: If $z \geqq z'$, go to Step 4; otherwise, set $z' = z$ and retain the corresponding stock levels as the incumbent stock levels. Go to Step 4.

Step 4: Increment g' by b. If $f > C - g'$, then terminate; otherwise, return to Step 1.

Let us now discuss the algorithm in greater detail. In Step 1 an optimization problem is solved in which a portion of the budget, g', is allocated among the modules to de-

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termine depot and base module stock levels. The objective of this subproblem is to determine that division of g' which minimizes the sum of the expected weighted delay functions. By substituting for Δ_i the objective function to (3) is

$$\min \sum_{j=1}^{N} [\sum_{i=1}^{M} \sum_{x_{ij}>s_{ij}} r_i(x_{ij} - s_{ij})p(x_{ij}|\lambda_{ij}T_{ij})].$$

Problem (3) is therefore separable with respect to modules. The solution to (3) is obtained by constructing the corresponding Lagrangian function and solving the following N module subproblems for a specific value of the Lagrangian multiplier, γ,

$$\min [\sum_{i=1}^{M} \sum_{x_{ij}>s_{ij}} r_i(x_{ij} - s_{ij})p(x_{ij}|\lambda_{ij}T_{ij}) - \gamma c_j \sum_{i=0}^{M} s_{ij}]. \tag{5}$$

Unfortunately, this problem is neither convex nor separable; however, the minimand is strictly decreasing in s_{ij} and s_{0j}, and for a fixed value of depot stock is discretely convex with respect to s_{ij}. Therefore, the solution to (5) is obtained by solving the following problem for $s_{0j} = 0, 1, \cdots$:

$$h(s_0) = \min \sum_{i=1}^{M} [\sum_{x_{ij}>s_{ij}} r_i(x_{ij} - s_{ij})p(x_{ij}|\lambda_{ij}T_{ij}) - \gamma c_j s_{ij}].$$

The optimal base module stock level is the first nonnegative integer s_{ij} satisfying

$$\sum_{x_{ij}=0}^{s_{ij}} p(x_{ij}|\lambda_{ij}T_{ij}) \geqq 1 + \gamma c_j/r_i .$$

The optimal solution to (5) corresponds to the depot stock level yielding the minimum value of $h(s_0) - \gamma c_j s_{0j}$.

The solution to (5) is found for a particular value of the Lagrangian multiplier, γ. A range of values is specified for the multiplier in advance consistent with the minimum module budget, g, that is $\gamma_0 \leqq \gamma < 0$. The total module cost is calculated for each value of γ that is explicitly examined. An adaptive bisection procedure is used to determine the multiplier value that most closely corresponds to the given budget constraint, g', beginning with γ_0. In the actual problems that were solved, from seven to ten multiplier values were explicitly examined to guarantee convergence within $\frac{1}{2}$% of the target budget. The actual number depends on the quantity of modules considered and the range of their unit prices. Each bisection corresponds to examining a specific value of γ. By performing ten bisections, 1024 multiplier values are implicitly examined.

The engine subproblem is solved in Step 2. The values of the average base engine resupply times are adjusted to reflect the expected delays which correspond to the module stock levels computed in Step 1. The optimal solution to Problem (4) is obtained using a brute force method and a marginal analysis technique. Problem (4) is neither separable nor convex; however, by partitioning this problem in the same manner as the module problem, a sequence of problems of the form

$$\min \sum_i \sum_{x_i>s_i} (x_i - s_i)p(x_i|\lambda_i T_i)$$

subject to

$$\sum_i c_E s_i \leqq C - g' - c_E s_0 , \qquad s_i = 0, 1, \cdots , \tag{6}$$

are solved, one for each value of s_0. Each time (6) is solved, the value of T_i is adjusted to reflect the value of s_0. The solution to (6) is found by calculating the greatest integer less than or equal to $(C - g' - c_E s_0)/c_E$. Denote this value by I. Then the constraint in (6) is replaced by $\sum s_i \leqq I$. For a given value of s_0, Problem (6) is convex. Therefore, the solution to (6) can be obtained using a simple marginal analysis technique. The optimal solution corresponds to the depot stock level yielding the lowest backorders.

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In both the solution to the module and engine subproblems, a brute force method was employed to find the optimal value of depot stock. This approach appears to be naive; but, it is easy to show that the explicit search can be restricted to a very limited number of values in both cases. In practice, this number rarely exceeds ten. The region of search can be confined to the region containing the depot mean demand, for once depot stock exceeds the depot mean, $\delta(s_0)$ (or $\delta(s_{0j})$) rapidly approaches 0. For example, suppose there are six bases stocking engines with $A_i = B_i = 20$, $\lambda_i = 0.1$, $r_i = 0.9$, $i = 1, \cdots, 6$ and $D = 40$. The values in Table I indicate how $\delta(s_0)$ and T change as a function of s_0. The mean depot demand, θ, is 0.6. Thus the mean of the Poisson distribution used to calculate $\delta(s_0)$ and T is $\theta D = 2.4$. Consequently, by taking advantage of the problem's mathematical structure, the required solutions can be found with minimal effort.

The search for the optimal splitting of the budget between the engine and modules can also be improved. First, the range of module budgets that need to be explicitly evaluated is limited. An upper limit for module investment is related to the marginal reduction in engine delay due to an increased allocation of dollars. This marginal reduction rapidly approaches zero as investment in modules increases. The upper limit can be chosen to correspond to the point at which this marginal reduction is less than some small positive number. The lower limit can be established in a similar manner by recognizing that excessive delays can be avoided by making a minimal investment in modules. The search can be confined to this region. In a similar fashion, bounds can easily be established for minimal investment in engines by establishing a system engine backorder constraint. The value of γ_0 can be chosen to reflect the minimum desired investment level in modules.

The search for the best division of the budget between engines and modules can be reduced by exploiting the apparent strict quasi-convexity of expected base backorders for engines as a function of investment in engines. Thus, using a simple procedure, such as Fibonacci search, substantially reduces the computational requirements.

An example of the engine/module mix problem is displayed in Figure 2. The figure shows how engine backorders vary as the portion of the total budget allocated to engines changes for a range of budgets. For example, for a total budget of 180 million dollars, the optimal portion of the budget that should be allocated to engines is about 120 million dollars. The sensitivity of the system backorders to the investment in engines is apparent. Substantial errors in the mix between engines and modules will cause unnecessarily high backorders. It is also interesting to note that an under-

TABLE 1

Changes in $\delta(s_0)$ and T as a Function of s_0

Depot Stock	$\delta(s) = \frac{\Sigma (x - s)p(x \mid \theta D)}{\theta D}$	T
0	1.000	24
1	0.621	22.844
2	0.333	21.333
3	0.154	20.616
4	0.062	20.246
5	0.022	20.088
6	0.0017	20.0067
8	0.0004	20.0016
9	0.0000	20.000

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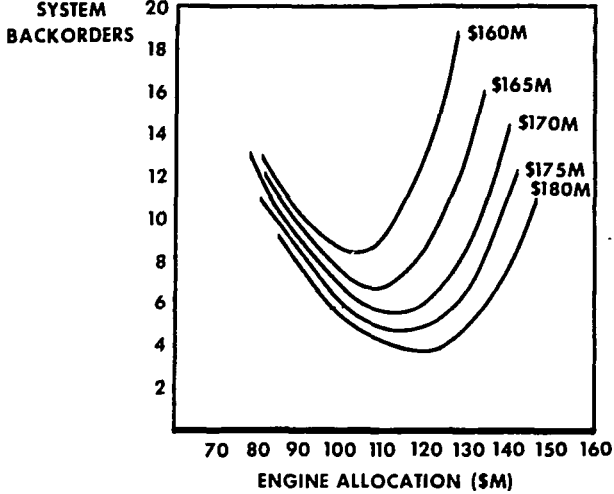

FIGURE 2. System backorders as a function of the portion of the total budget allocated to engines.

investment in engines produces less of an increase in backorders than a like over-investment.

The objective of the module optimization subproblem discussed above is to minimize $\sum_i r_i\lambda_i\Delta_i$. This is really an approximation. To guarantee an optimal solution, the objective of this problem should be to measure the direct impact on engine backorders of making an allocation of dollars among the modules. This requires the calculation of the marginal impact on the state probabilities and ultimately system backorders. The computational difficulties involved in this approach are obvious. However, for reasonable module budgets, the reduction in expected engine base backorders is closely related to the reduction in the $r_i\lambda_i\Delta_i$. In practice, the "tail" probabilities used in the calculations of expected engine backorders are very similar between bases. Furthermore, for the budgets considered in most practical situations these are the only probabilities of interest. Since the backorders change in about the same way that the $r_i\lambda_i\Delta_i$ change, the effect of using this approximation is unlikely to alter the optimal solution. For most planning situations there are, at most, a few different classes of bases, e.g., training, tactical, and overseas bases that need to be considered. The approximation seems to be quite reasonable for these problems.

6. Summary and Concluding Comments

The first portion of this paper was devoted to a review of the mathematical inventory model called METRIC. Subsequently, a mathematical model, which is an extension of METRIC, was presented for controlling inventory in a multi-item, multi-echelon, multi-indenture system. An example application of the model relating to the calculation of requirements for spare aircraft engines and modules was discussed. Finally, a solution algorithm for obtaining depot and base stock levels was presented.

It is easy to see that the concepts discussed in this paper can be used to determine stock levels for a collection of repairable assemblies and their repairable subordinate components. Define a line replacement unit (LRU) to be a recoverable item for which

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the maintenance concept is to remove the item from the aircraft when it fails and to replace it with a similar item. A shop replaceable unit (SRU) is one which is removed from an LRU and replaced in the LRU in a maintenance shop. An engine and its modules are examples of an LRU and SRUs, respectively. In the MOD-METRIC model and algorithm, attention was restricted to a single LRU and its subordinate SRUs (LRU group). However, by employing a simple marginal analysis technique, it is possible to compute stock levels for many LRUs and their subordinate SRUs.

To accomplish this task, it is necessary to partition the overall problem to the LRU level and to perform a MOD-METRIC analysis on each LRU group. Once this is accomplished, a piece-wise linear performance/cost functional relationship can be constructed for each LRU group. If this function is not convex, it must be replaced by its greatest convex minorant. Using marginal analysis these individual performance/cost functions are used to obtain a solution to the multi-LRU problem. Therefore, the model can be extended to solve problems in which many LRU/SRU groups are considered simultaneously. The objective in this extended model is to minimize total expected base LRU backorders subject to a constraint on investment in all LRUs and SRUs. This model is the one the Air Force has implemented as the method for computing recoverable spare stock levels for the F-15 weapon system.

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